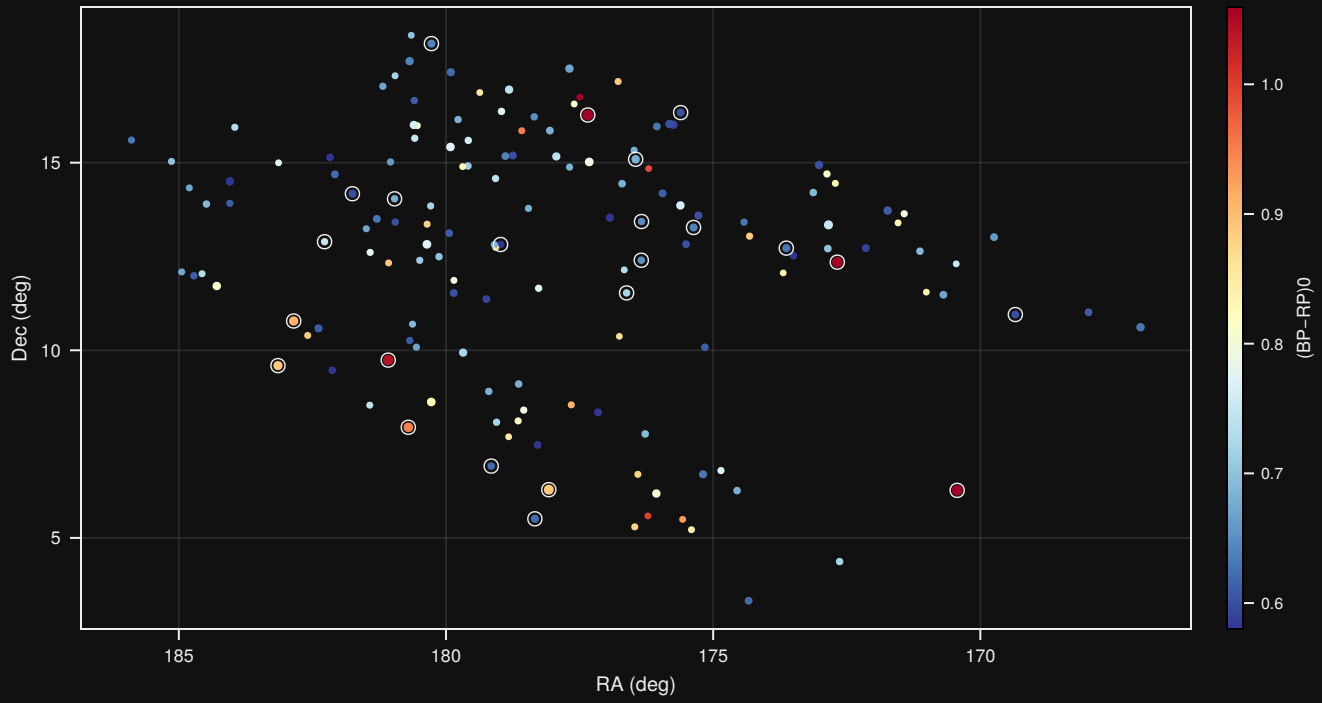
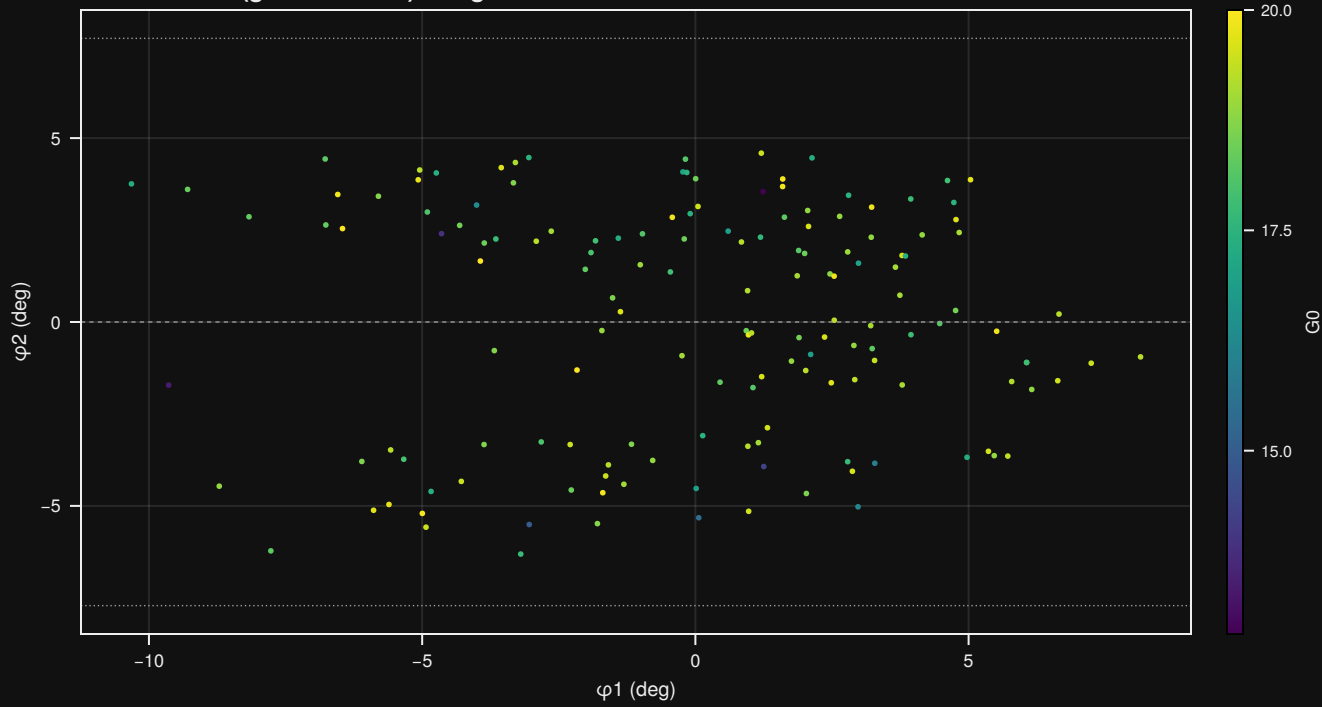
New-15

MEMBERSHIP	
Stars in catalog	152
with Gaia GSP-Phot	80 (53%)
with radial velocity	23
Stream id (sID)	43
GEOMETRY · great-circle fit	
Centroid RA, Dec	178.0, 13.1 deg
Centroid l, b	256.6, 70.4 deg
Pole RA, Dec	66.8, 58.4 deg
Length (φ1, 2–98%)	15.3 deg
Width σ(φ2), robust	3.86 deg
KINEMATICS	
pmRA* med ± σ	3.70 ± 0.68 mas/yr
pmDE med ± σ	-3.47 ± 0.69 mas/yr
μφ1* med ± σ	1.58 ± 0.46 mas/yr
μφ2 med ± σ	-4.89 ± 0.80 mas/yr
Parallax (cat.) med	0.146 ± 0.231 mas
Parallax (wtd, ZP-corr.)	0.173 ± 0.014 mas
Distance	d = 5.8 ± 0.5 kpc
Physical length	1.55 kpc
RV median ± σ(MAD)	191.6 ± 14.7 km/s
RV gradient dV/dφ1	-0.18 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.9–20.0 (18.8)
(BP−RP)0 median	0.69
GSP-Phot [M/H] (n=80)	-2.18 ± 0.95
GSP-Phot Teff median	6094 K
RUWE > 1.4	0.0 %
PM clusters (largest)	10 (17 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	152 | 152
n_RV catalog | paper	23 | 23
<[Fe/H]> paper	-2.01
Stellar mass (paper)	960 Msun
Pole offset vs paper	0.7 deg

Sky map · catalog members (n = 152) · size ∝ brightness · ring = has RV

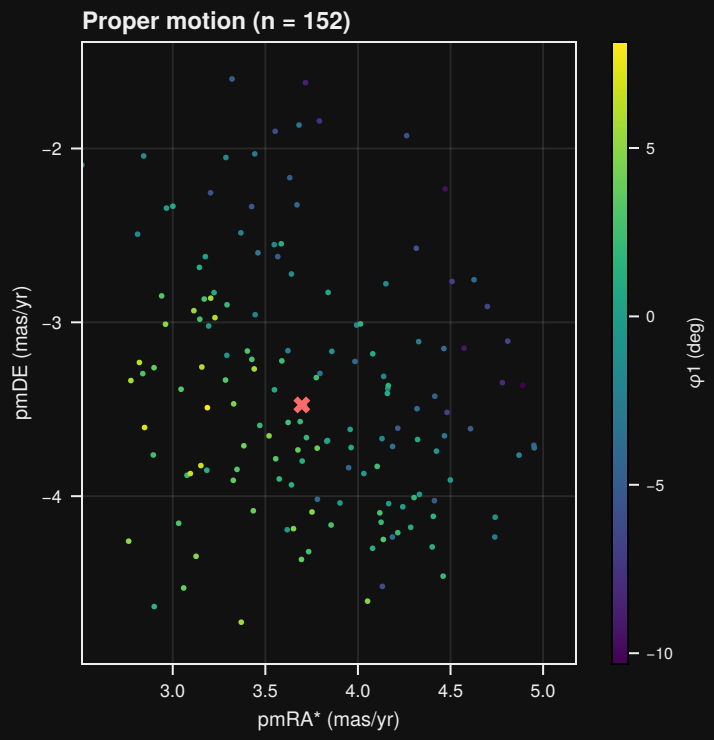


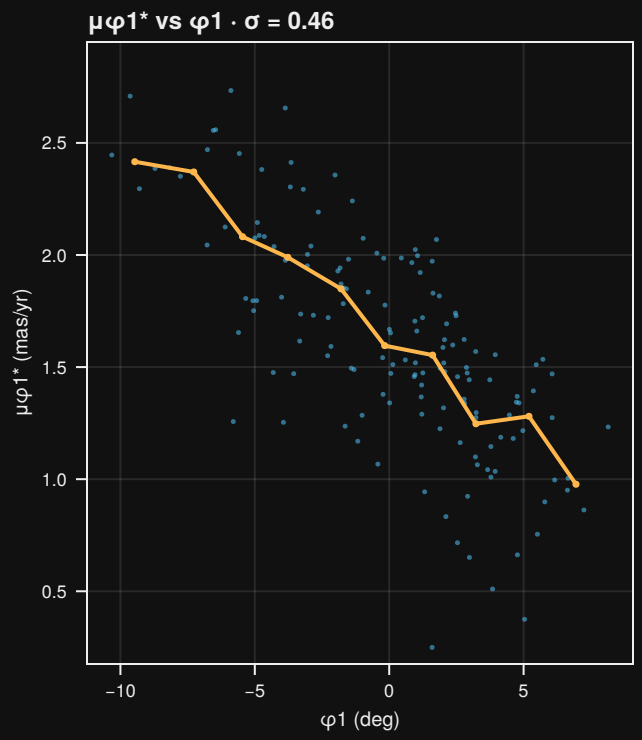
Stream frame (great-circle fit) · length 15.3° · width σ 3.86°

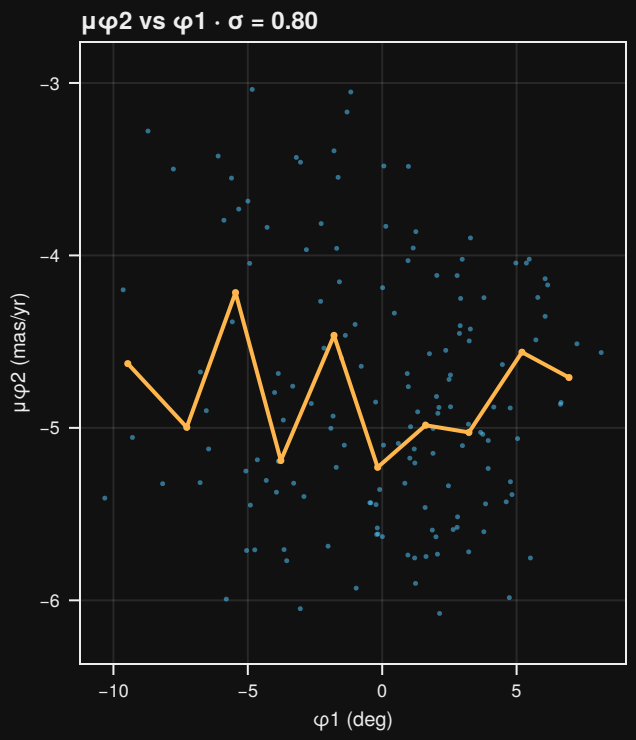


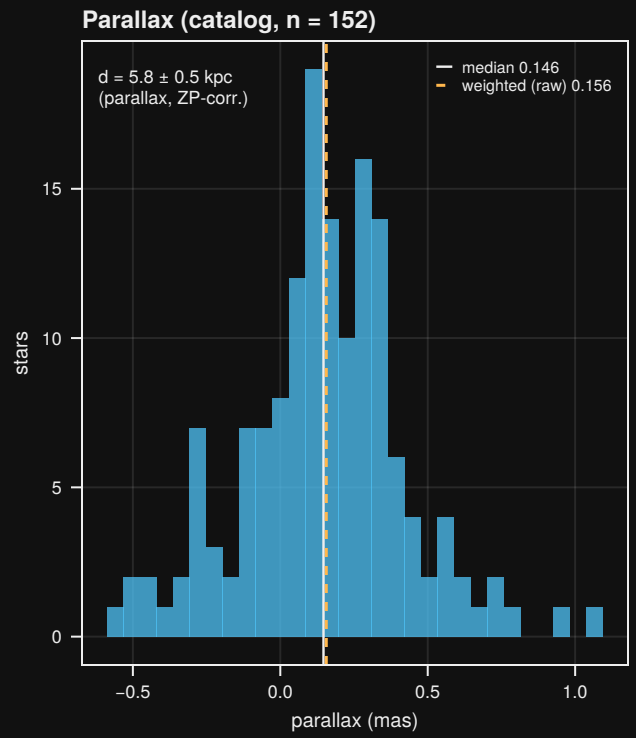
New-15

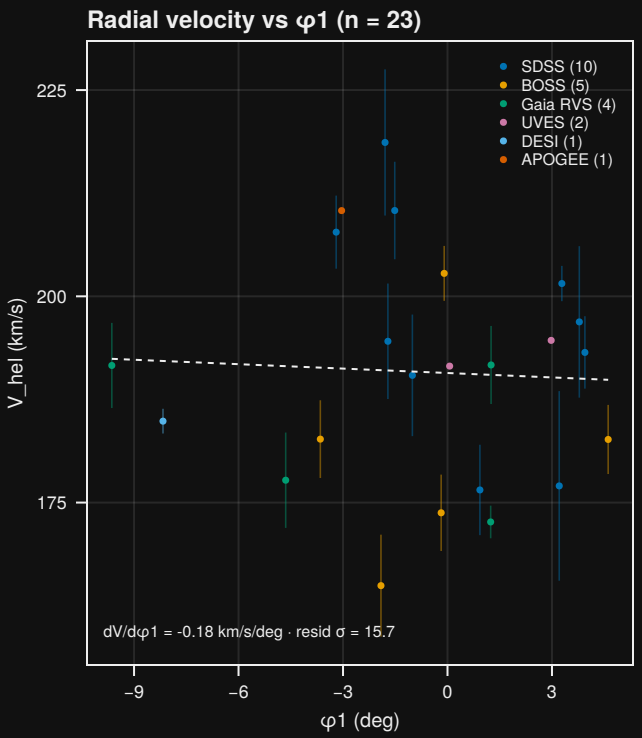
Kinematics · proper motion · parallax · radial velocity

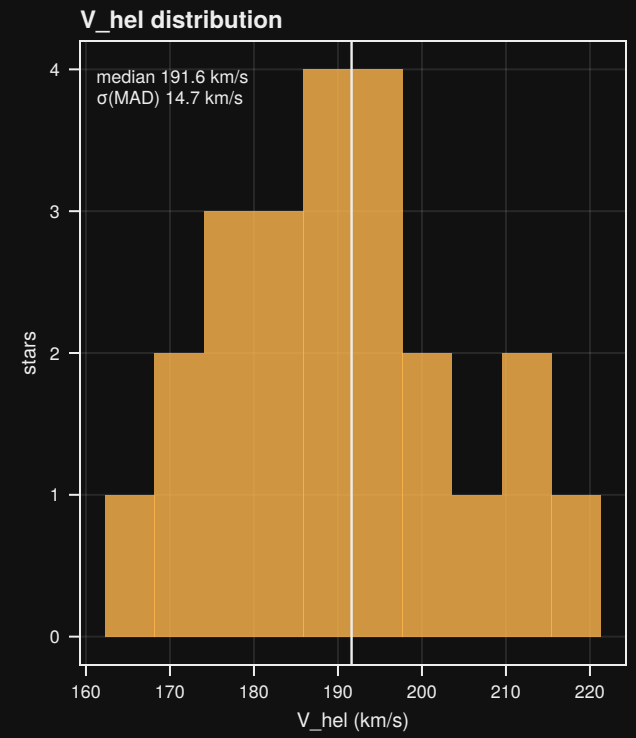






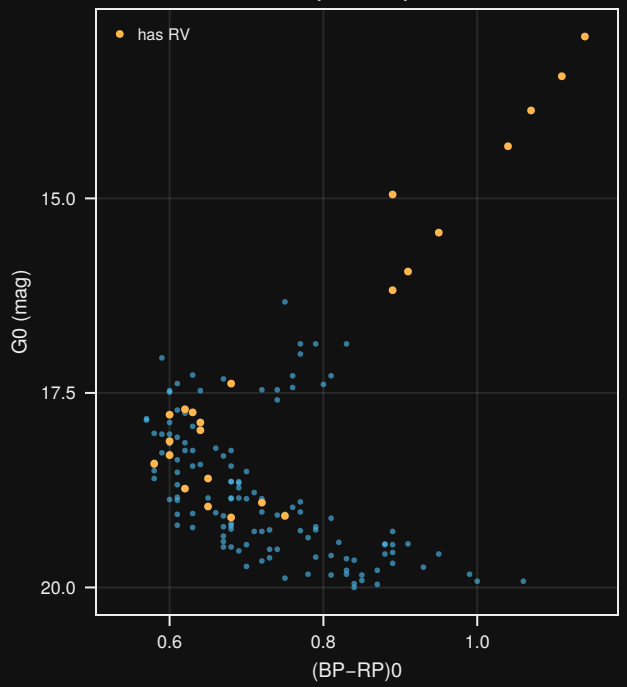




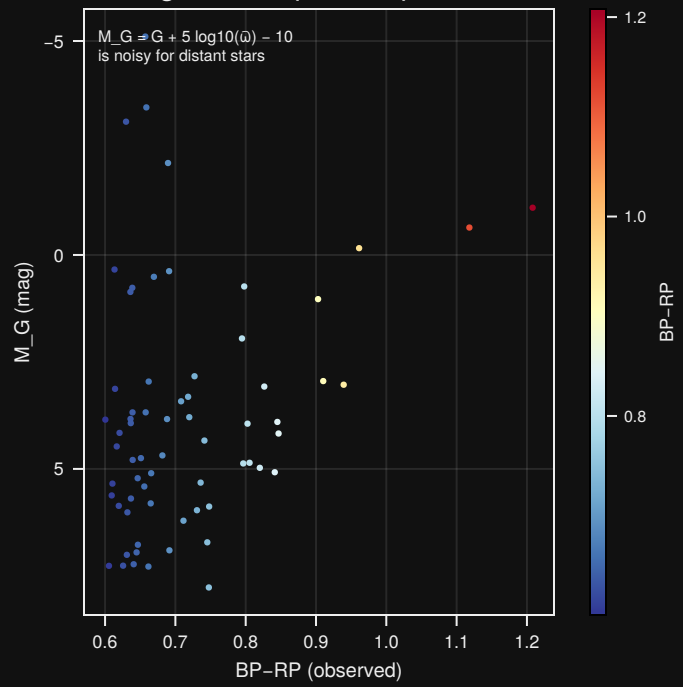


New-15

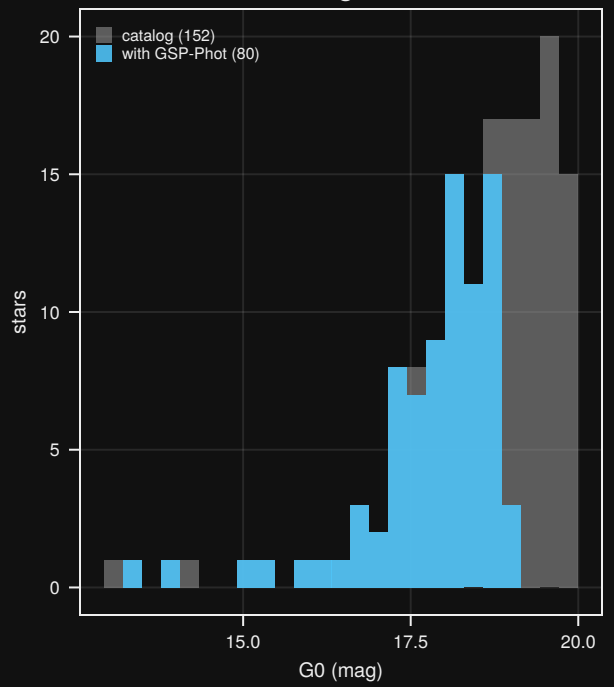
CMD · dereddened (n = 152)



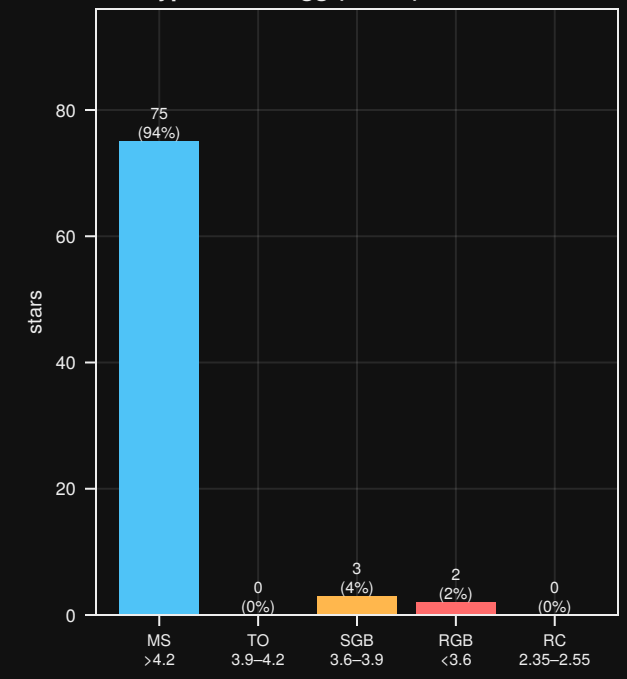
HR diagram · ϖ>0 (n = 66/80)



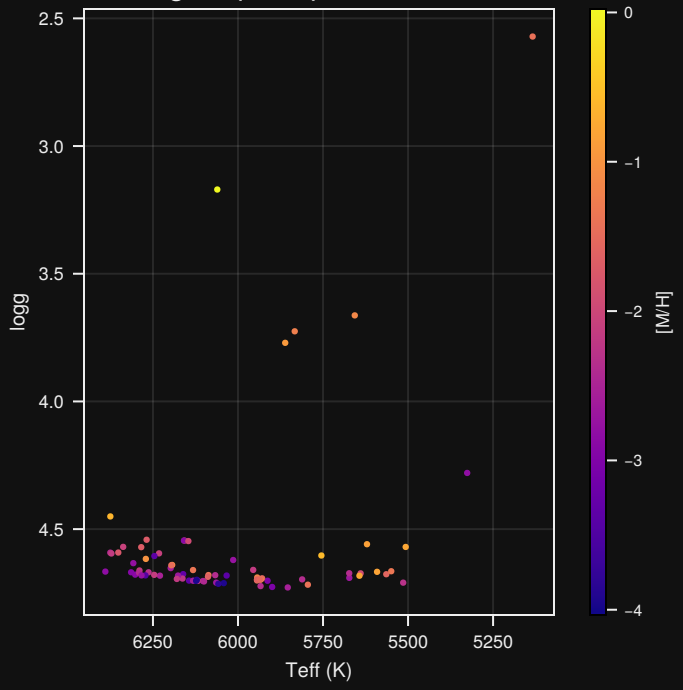
G0 & GSP-Phot coverage



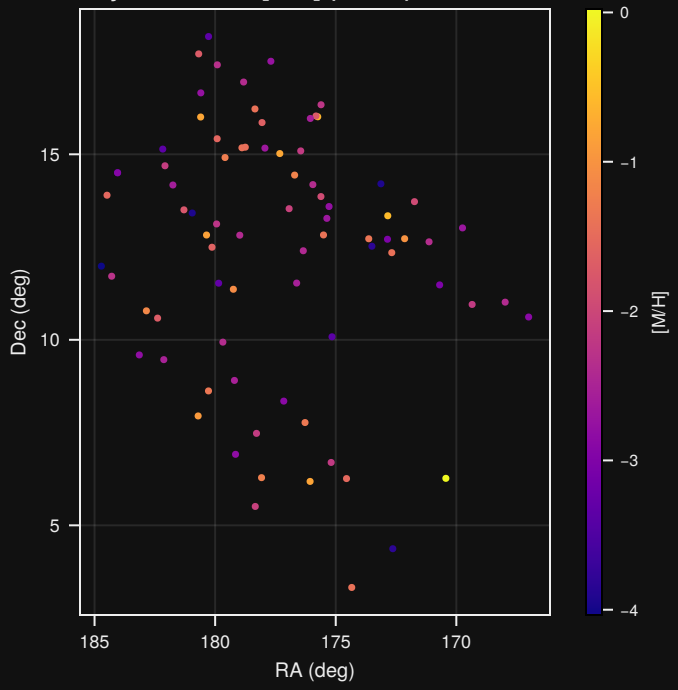
Star types from logg (n = 80)



Kiel diagram (n = 80)



Sky · GSP-Phot [M/H] (n = 80)

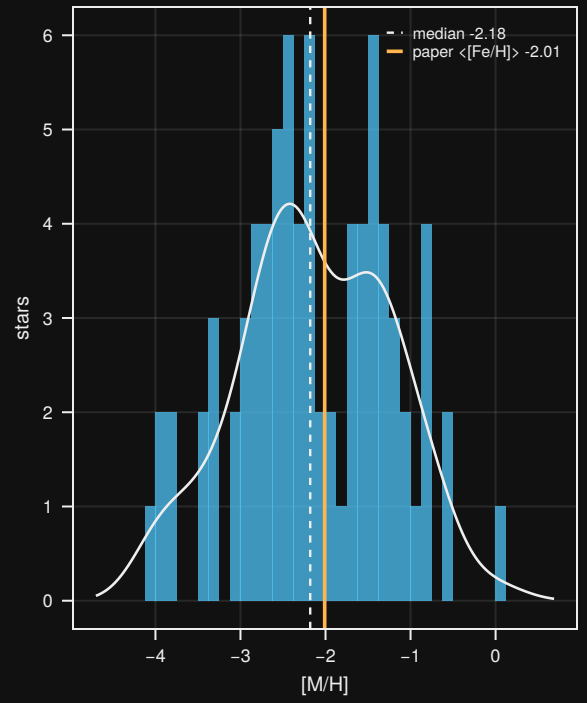


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

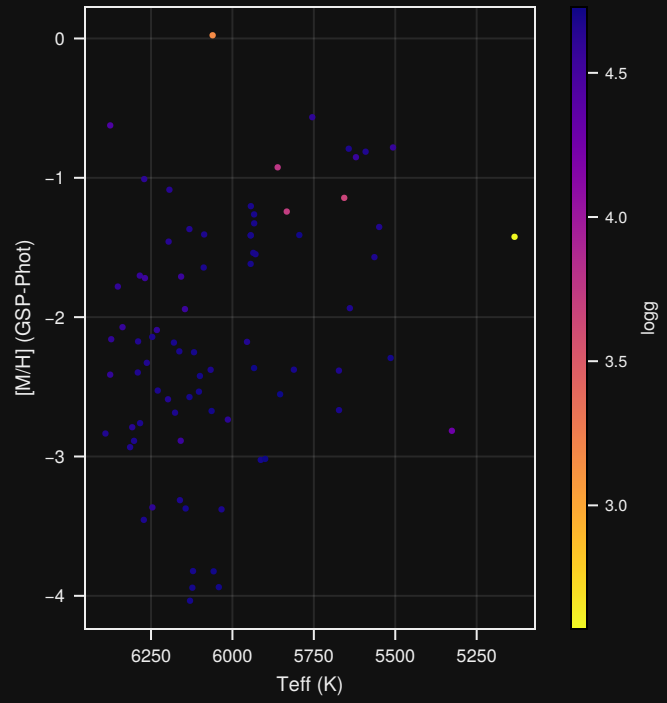
New-15

Metallicity · age · astrometric quality

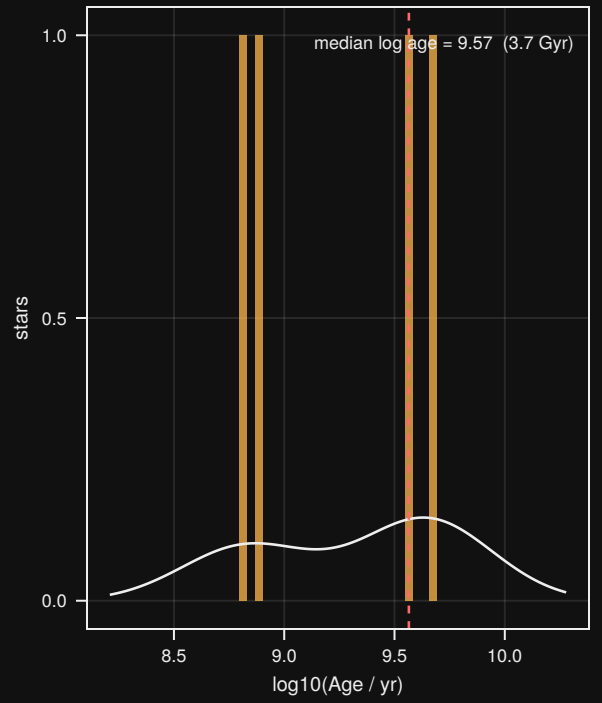
[M/H] GSP-Phot (n = 80)



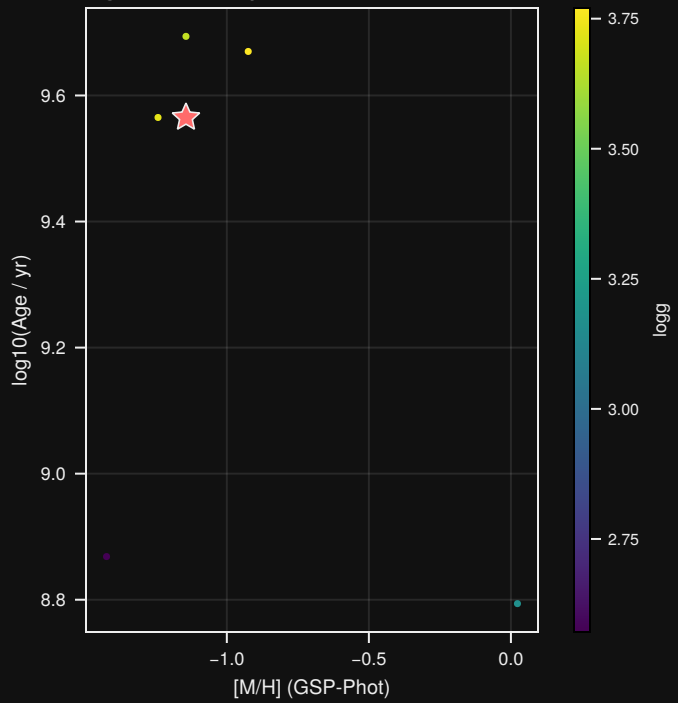
Teff vs [M/H] (n = 80)



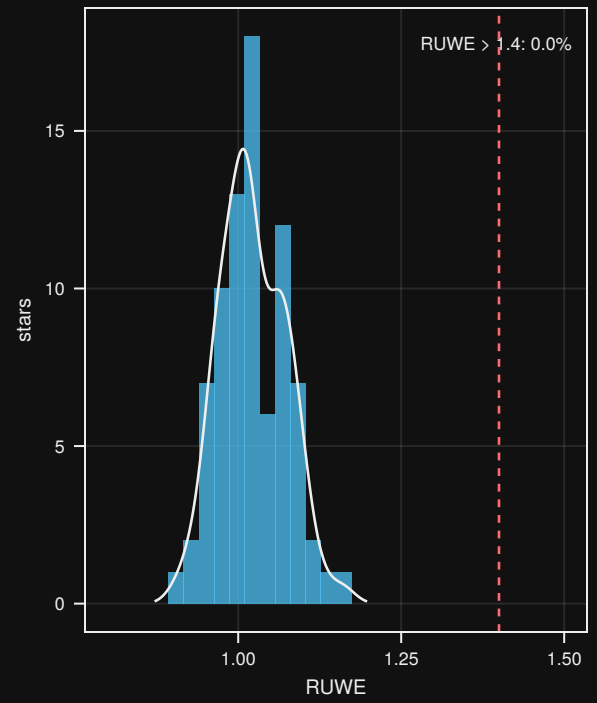
FLAME age (n = 5)



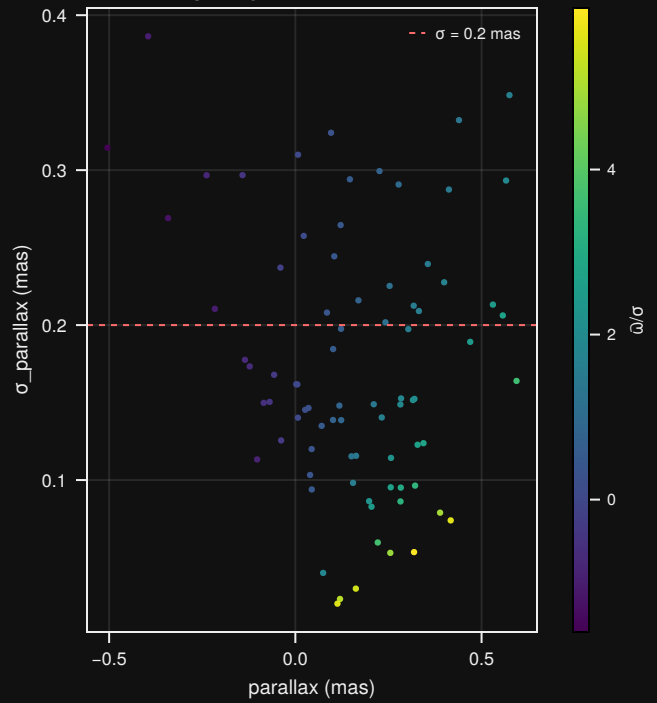
Age–metallicity (n = 5)



RUWE (n = 80)

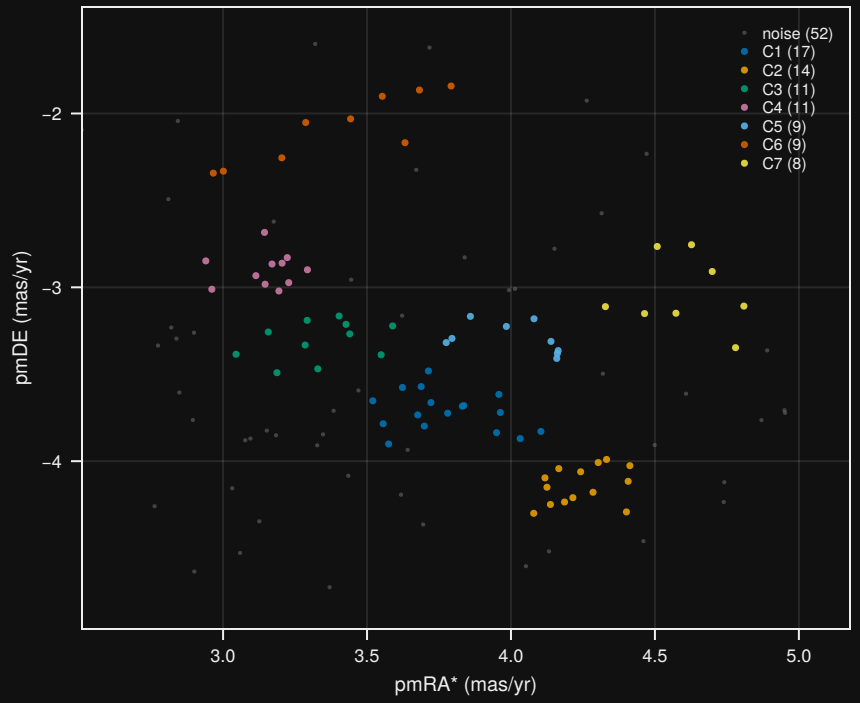


Parallax quality (n = 80)

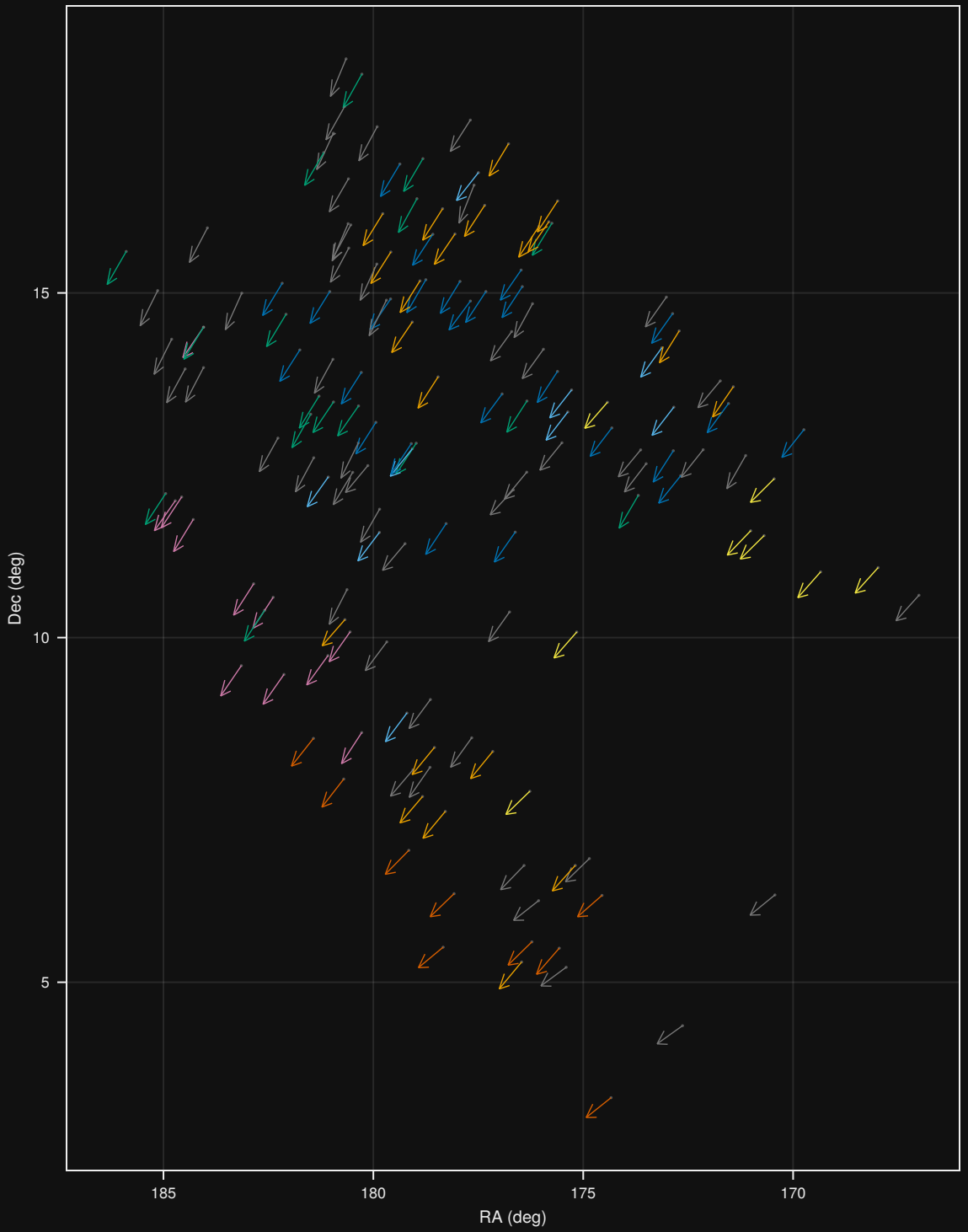


New-15

PM clustering · HDBSCAN · 10 cluster(s) · noise 34%



Proper-motion directions (arrows) coloured by cluster



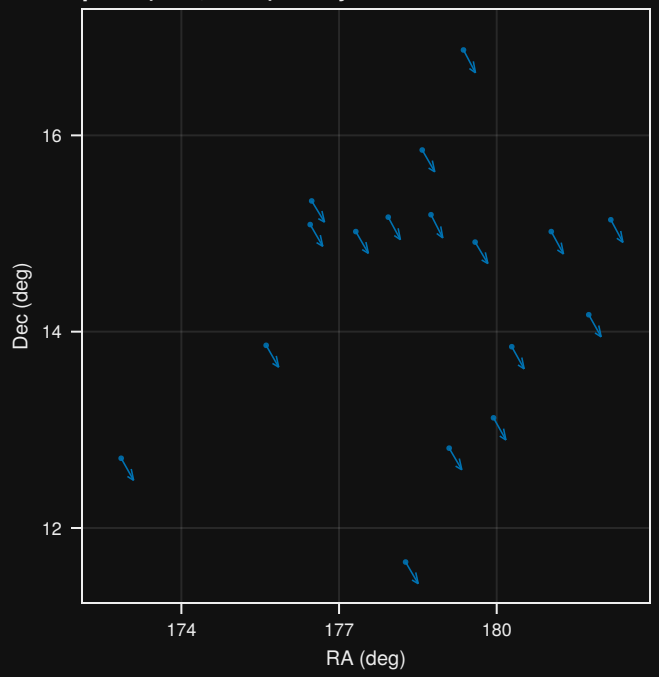
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	52	3.46	-3.71	0.83	-1.66	5
C1	17	3.72	-3.72	0.17	-2.21	2
C2	14	4.23	-4.13	0.14	-1.68	2
C3	11	3.33	-3.27	0.16	-2.41	2
C4	11	3.17	-2.90	0.08	-2.41	3
C5	9	4.08	-3.31	0.12	-2.73	1
C6	9	3.44	-2.05	0.35	-1.48	4
C7	8	4.60	-3.11	0.17	-2.40	1
C8	8	4.37	-3.66	0.16	-2.31	2

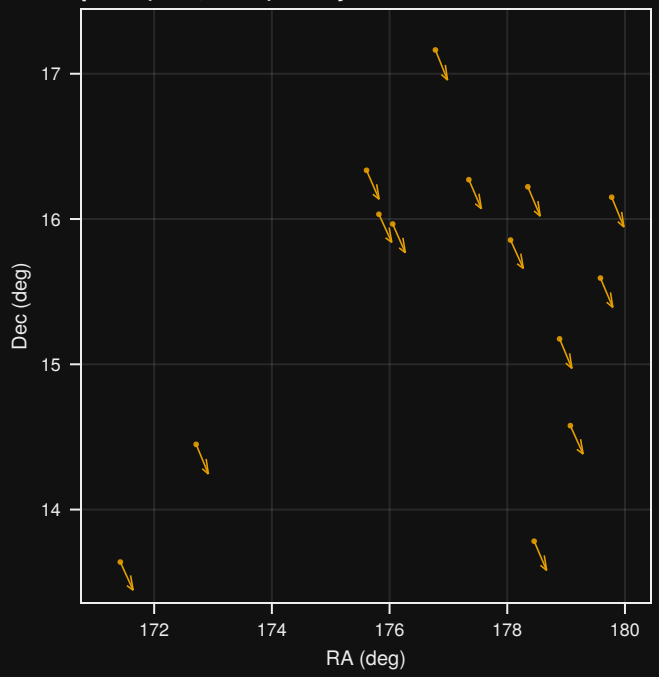
New-15

Per-cluster proper-motion vector maps

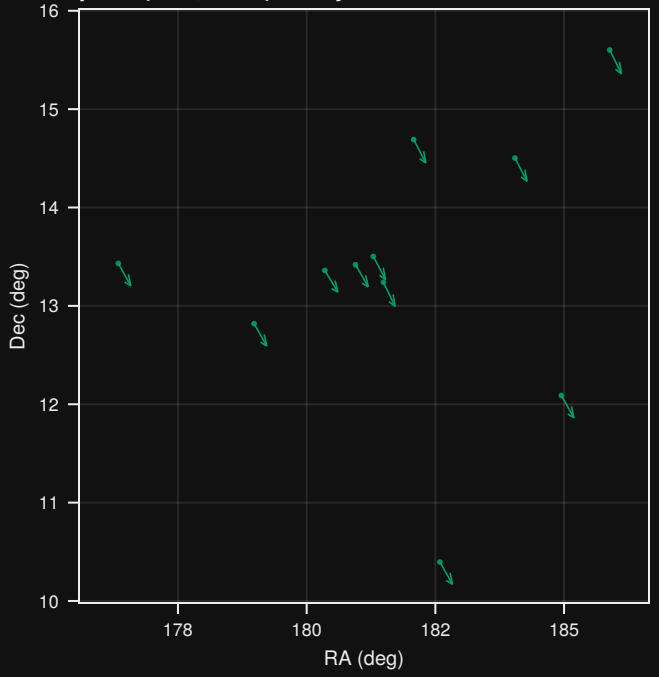
Cluster C1 · n = 17
pm = (3.72, -3.72) mas/yr



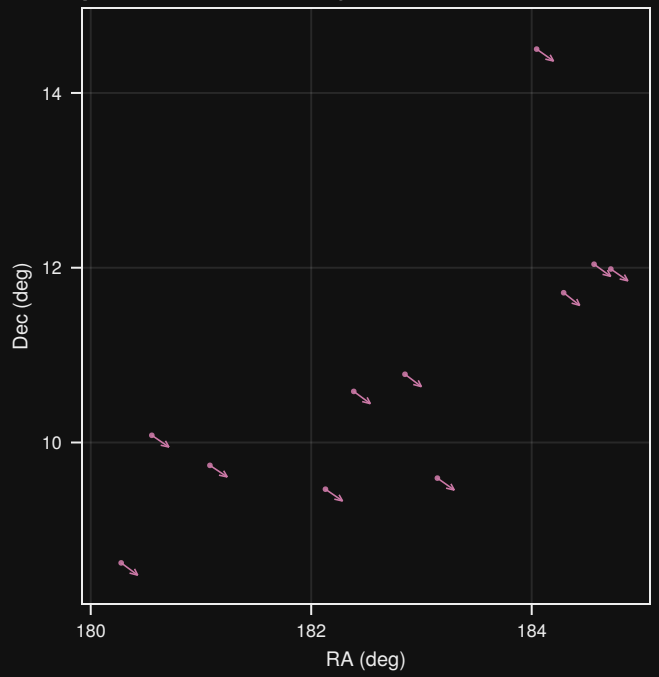
Cluster C2 · n = 14
pm = (4.23, -4.13) mas/yr



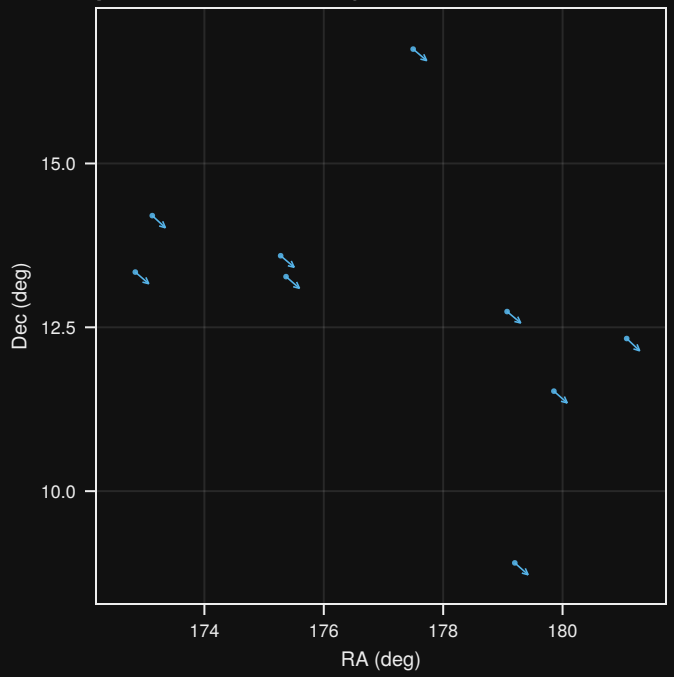
Cluster C3 · n = 11
pm = (3.33, -3.27) mas/yr



Cluster C4 · n = 11
pm = (3.17, -2.90) mas/yr



Cluster C5 · n = 9
pm = (4.08, -3.31) mas/yr



Noise · n = 52

